Yuan YIN 银元

AI Researcher · Model adaptation, learned dynamics, autonomous driving

yuan-yin.github.io • Google Scholar • yuan-yin-nn • yuan-yin • Paris, France

Profile

AI researcher working on **parameter-efficient adaptation of pretrained vision models** and **learned physical dynamics**, applied to autonomous driving. My PhD established hybrid physics/deep-learning modeling and out-of-distribution generalization for dynamical systems; I now transfer that work to autonomous driving at an open industrial research lab. I favor methods that exploit the structure of a problem and I work on integrating deep learning into existing non-ML systems.

Experience

Valeo.ai *Paris, France*

- **AI Researcher** Dec 2024 – Present
  Perception, motion forecasting and decision making for autonomous driving; adapting large pretrained models and learning at scale
- **Postdoctoral AI Researcher** Apr 2024 – Nov 2024
  Corner-case generation for learning robust self-driving cars

Sorbonne Université *ISIR, MLIA* *Paris, France*

- **Postdoctoral Researcher** Jul 2023 – Dec 2023
  Supervised ongoing research projects; authored a tutorial on physics-aware deep learning
- **PhD Student & Teaching Assistant** Oct 2019 – Jun 2023
  Advisors: Patrick Gallinari & Nicolas Baskiotis
  Physics-aware deep learning for dynamical systems: hybrid physics/DL modeling, out-of-distribution generalization, continuous-time dynamics
- **Research Intern, Deep Learning** Feb 2019 – Sep 2019
  Imputing spatiotemporal data with generative models

Inria Paris *Research Intern, NLP* 2018

Beihang Univ. *Research Intern, Vision* 2015–16

Education

Sorbonne Université fka UPMC (Paris-6) *Paris, France*

- **PhD**, Machine Learning & Deep Learning 2023
- **MSc2 DAC** *Master Data Science Paris* 2019

Université Paris Cité fka Paris-Diderot (Paris-7) *Paris, France*

- **MSc1 MPRI** *Parisian Research Master in CS* 2018
- Univ. Diploma, French Language & Civilization 2017

Beihang University *Beijing, China*

- **BSc**, Computer Science 2016

Technical Skills

ML / DL	PyTorch, JAX, Hugging Face, multi-GPU & multi-node training
Coding	Python, C/C++, Java, OCaml, Matlab, LaTeX
Infra	Linux, Slurm, Docker, Git, Weights & Biases

Languages

French	Bilingual	*last exam ▷ C1 (2017)*
English	Full Professional	*last exam ▷ B2 (2015)*
Mandarin	Native	

Awards & Honors

Accessit for the 2024 AI Thesis Prize of the French Association for Artificial Intelligence (AFIA)

Best Paper & Oral, NeurIPS 2025 CCFM Workshop

Top Reviewer, NeurIPS 2023

Service & Teaching

Conference reviewing — NeurIPS 2021–26, ICLR 2023–26, ICML 2022–26, CVPR 2025–26, ICRA 2026, ECML-PKDD 2021, ACM-MM 2021

Workshop reviewing — CCFM @ NeurIPS 2025, ML4PS @ NeurIPS 2022–24, Physics4ML @ ICLR 2023, SynS&ML @ ICML 2023, ROAM @ ECCV 2024

Teaching — Sorbonne Université, Engineering Dept. (UFR 919), 2019–22, in French. Undergraduate: C programming (L1), algorithmics (L2), probability (L3). Graduate: ML research methodology (M2)

Selected Publications

Conference & journal papers * equal contribution

- E. Kirby, A. Boulch, Y.-H. Xu, **Y. Yin**, G. Puy, É. Zablocki, A. Bursuc, S. Gidaris, R. Marlet, F. Bartoccioni, A.-Q. Cao, N. Samet, T.-H. Vu, and M. Cord. Driving on registers. In ***CVPR 2026***.
- Y.-H. Xu*, **Y. Yin***, T.-H. Vu, A. Boulch, É. Zablocki, and M. Cord. PPT: Pre-training with pseudo-labeled trajectories for motion

forecasting. In ***ICRA 2026***.

- **Y. Yin**, S. Venkataramanan, T.-H. Vu, A. Bursuc, and M. Cord. IPA: An information-reconstructive input projection framework for efficient foundation model adaptation. ***TMLR***, 2025. **Best Paper & Oral, NeurIPS 2025 CCFM WS**
- L. Le Boudec, E. de Bézenac, L. Serrano, R. D. Regueiro-Espino, **Y. Yin**, and P. Gallinari. Learning a neural solver for parametric PDE to enhance physics-informed methods. In ***ICLR 2025***.
- A. Kassaï Koupaï, J. Mifsut-Benet, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. GEPS: Boosting generalization in parametric PDE neural solvers through adaptive conditioning. In ***NeurIPS 2024***.
- E. Le Naour, L. Serrano, L. Migus, **Y. Yin**, G. Agoua, N. Baskiotis, P. Gallinari, and V. Guigue. Time series continuous modeling for imputation and forecasting with implicit neural representations. ***TMLR***, 2024.
- L. Serrano, L. Le Boudec, A. Kassaï Koupaï, T. X. Wang, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. Operator learning with neural fields: Tackling PDEs on general geometries. In ***NeurIPS 2023***.
- **Y. Yin***, M. Kirchmeyer*, J.-Y. Franceschi*, A. Rakotomamonjy, and P. Gallinari. Continuous PDE dynamics forecasting with implicit neural representations. In ***ICLR 2023***. **Spotlight**
- M. Kirchmeyer*, **Y. Yin***, J. Donà, N. Baskiotis, A. Rakotomamonjy, and P. Gallinari. Generalizing to new physical systems via context-informed dynamics model. In ***ICML 2022***. **Spotlight**
- **Y. Yin**, I. Ayed, E. de Bézenac, N. Baskiotis, and P. Gallinari. LEADS: Learning dynamical systems that generalize across environments. In ***NeurIPS 2021***.
- **Y. Yin***, V. Le Guen*, J. Donà*, E. de Bézenac*, I. Ayed*, N. Thome, and P. Gallinari. Augmenting physical models with deep networks for complex dynamics forecasting. In ***ICLR 2021***. **Oral; also in J. Stat. Mech. Theory Exp.**
- D. Huang, R.K. Zhang, **Y. Yin**, Y.D. Wang, and Y.H. Wang. Local feature approach to dorsal hand vein recognition by centroid-based circular key-point grid and fine-grained matching. ***Image Vis. Comput.***, 2017.

Workshop papers

- **Y. Yin**, P. Khayatan, É. Zablocki, A. Boulch, and M. Cord. ReGentS: Real-world safety-critical driving scenario generation made stable. In ***ECCV 2024*** *Workshop on W-CODA*.
- L. Serrano, L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. INFINITY: Neural field modeling for reynolds-averaged navier-stokes equations. In ***ICML 2023*** *Workshop on SynS & ML*.
- L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. Multi-scale physical representations for approximating PDE solutions with graph neural operators. In ***ICLR 2022*** *Workshop on GTRL*.
- **Y. Yin**, A. Pajot, E. De Bézenac, and P. Gallinari. Unsupervised inpainting for occluded sea surface temperature sequences. In ***CI 2019***.

Preprints

- **Y. Yin**, E. Ramzi, M. Lafon, V. Charraut, V. Bares, Y.-H. Xu, É. Zablocki, A. Boulch, T. Buhet, A. Bursuc, and M. Cord. Pictura: Perspective-view self-play at scale for driving. **arXiv** 2607.26005, 2026.
- Y.-H. Xu, É. Zablocki, **Y. Yin**, E. Ramzi, E. Kirby, A. Boulch, and M. Cord. TOAD: Test-time trajectory optimization for autonomous driving. **arXiv** 2606.07170, 2026.

Full list of publications on Google Scholar

Selected Talks

Oral, NeurIPS 2025 CCFM Workshop	Dec 2025
Poster, ECCV 2024	Sep 2024
Workshop on Mathematical Foundations of AI, DATAIA–SCAI	Jan 2024
Tutorial, ECML-PKDD 2023	Sep 2023
Seminar, Signal Processing Lab (LTS4), EPFL	May 2023
Spotlight, ICLR 2023	May 2023
AI4Science Talks, Univ. of Stuttgart & NEC Labs Europe	Apr 2023
Seminar, Criteo AI Lab	Nov 2022
Spotlight, ICML 2022	Jul 2022

Full list of talks on my website